

Galois Theory - MM7043 Autumn 2026

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1 Fields

Throughout the semester, we will employ the following conventions:

- Unless otherwise specified, all rings will be assumed to be unital and commutative.
- A *ring homomorphism* is a map of rings $\varphi : R \rightarrow S$, such that
 1. $\varphi(x + y) = \varphi(x) + \varphi(y)$, for all $x, y \in R$.
 2. $\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y)$, for all $x, y \in R$.
 3. $\varphi(1_R) = 1_S$

Definition 1.1. Let R be a ring. An element $x \in R$ is called a *unit* or *invertible*, if there exists an element $y \in R$ such that $x \cdot y = 1$.

The set of all units of R form a group under multiplication and will be denoted R^* .

Example 1.2.

1. $\mathbb{Z}^* = \{-1, 1\}$
2. $\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$
3. For $n \in \mathbb{N}$, it holds that $(\mathbb{Z}_n)^* = \{m \in \mathbb{Z}_n : \gcd(m, n) = 1\}$.

Definition 1.3. A *field* is a ring $\mathbb{K}$ such that $\mathbb{K}^* = \mathbb{K} \setminus \{0\}$.

Example 1.4.

1. $\mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$
2. $\mathbb{Z}_p$, where $p \in \mathbb{P}$.

Lemma 1.5. A ring R is a field if and only if the only ideals of R are $\{0\}$ and R .

Proof. Let $I \subseteq R$ be an ideal. If I contains a unit then $I = R$.

“ $\implies$ ”: If $I \subseteq R$ is a non-zero, non-trivial ideal, then I contains a unit, which implies $I = R$.

“ $\impliedby$ ”: If R is not a field, then there exists some non-zero, non-unit element $x \in R$. But then the ideal Rx satisfies

$$\{0\} \subsetneq Rx \subsetneq R. \quad \square$$

Corollary 1.6. *A homomorphism of fields is always injective.*

Proof. Let $\varphi : \mathbb{K} \rightarrow \mathbb{L}$ be a field homomorphism. Then $\ker \varphi$ is an ideal in $\mathbb{K}$. We have $1_{\mathbb{K}} \notin \ker \varphi$, therefore $\ker \varphi = \{0\}$ and φ is injective. $\square$

We can construct new fields by considering subfields generated by a subset of a field.

Lemma 1.7. *Suppose $\mathbb{K}$ is a field and $\{\mathbb{F}_{\alpha} : \alpha \in I\}$ is a family of subfields of $\mathbb{K}$. Then*

$$\bigcap_{\alpha \in I} \mathbb{F}_{\alpha}$$

is a subfield of $\mathbb{K}$.

Definition 1.8. Suppose $\mathbb{L}_0 \subseteq \mathbb{L}$ are fields and $\alpha_1, \alpha_2, \dots \in \mathbb{L}$. Then we define $\mathbb{L}_0(\alpha_1, \alpha_2, \dots)$ as the intersection of all subfields of $\mathbb{L}$ that contain $\mathbb{L}_0$ and $\alpha_1, \alpha_2, \dots$, and will also be called the *extension* of $\mathbb{L}_0$ generated by $\alpha_1, \alpha_2, \dots$.

It is clear that such an extension is the smallest subfield of $\mathbb{L}$ that contains $\mathbb{L}_0$ and $\alpha_1, \alpha_2, \dots$.

Example 1.9.

1. Consider the fields $\mathbb{Q} \subseteq \mathbb{R}$, then one can show

$$\mathbb{Q}(\sqrt{2}) = \{p + q\sqrt{2} : p, q \in \mathbb{Q}\}.$$

2. Since the element $1_{\mathbb{Q}}$ already generates $\mathbb{Q}$ as a field, it follows that $\mathbb{Q}$ has no proper subfields, i.e. $\mathbb{Q}$ is a *prime field*.
3. The field $\mathbb{F}_p$ is a primary field for any $p \in \mathbb{P}$.

Definition 1.10. Let $\mathbb{F}$ be a field. The *primary subfield* of $\mathbb{F}$ is the subfield generated by the element $1_{\mathbb{F}}$.

Definition 1.11. Let R be a ring, then its *characteristic* is defined by

$$\text{char } R := \begin{cases} \min\{n \in \mathbb{N} : \sum_{j=1}^n 1_R = 0\}, & \text{if it exists,} \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 1.12. Let $\mathbb{F}$ be a field.

- If $\text{char } \mathbb{F} = 0$, then the primary subfield of $\mathbb{F}$ is¹ $\mathbb{Q}$.
- If $\text{char } \mathbb{F} = p \in \mathbb{P}$, then the primary subfield of $\mathbb{F}$ is $\mathbb{F}_p$.

Lemma 1.13. Suppose $M \subseteq R$ is a maximal ideal. Then the quotient R/M is a field.

Remark 1.14. Suppose $\mathbb{K}$ is a field and consider $\mathbb{K}[x]$, the ring of polynomials in x with coefficients in $\mathbb{K}$. Recall that $\mathbb{K}[x]$ is a principle ideal domain. A non-trivial ideal $(f(x))$ is maximal in $\mathbb{K}[x]$ if and only if $f(x)$ is an irreducible polynomial. In particular, $\mathbb{K}[x]/(f(x))$ is a field whenever $f(x)$ is irreducible over $\mathbb{K}$.

Moreover, $\mathbb{K}[x]/(f(x))$ contains $\mathbb{K}$ as a subfield, via the homomorphisms

$$\mathbb{K} \hookrightarrow \mathbb{K}[x] \rightarrow \mathbb{K}[x]/(f(x)).$$

Therefore, for every irreducible $f(x)$, $\mathbb{K}[x]/(f(x))$ is a field extension of $\mathbb{K}$.

Note that $f(x)$ being irreducible with $\deg f > 1$ implies that $f(x)$ has no roots in $\mathbb{K}$. But $f(x)$ always has a root in $\mathbb{K}[x]/(f(x))$, because $f(x)$ is zero as an element of $\mathbb{K}[x]/(f(x))$.

The above demonstrates (roughly) that there are relationships:

$$\begin{array}{ccc} \text{Polynomials} & \longleftrightarrow & \text{Field extensions} \\ \text{Properties of roots} & \longleftrightarrow & \text{Symmetries of field extensions} \end{array}$$

Practically, we want some criteria to determine if a polynomial is irreducible.

Remark 1.15. If $\mathbb{K}$ is a field and $p(x) \in \mathbb{K}[x]$, $\deg p > 1$ is a polynomial with a root in $\mathbb{K}$, then p is not irreducible. Indeed, for any $a \in \mathbb{K}$ one can write

$$p(x) = q(x)(x - a) + p(a),$$

for some polynomial $q(x) \in \mathbb{K}[x]$. If a is a root of $p(x)$, this demonstrates that $p(x)$ is not irreducible.

Note that the converse fails: The polynomial $(x^2 + 1)^2$ has no roots in $\mathbb{R}$ and is not irreducible.

¹We always mean “up to isomorphism”.

Lemma 1.16. *For a field $\mathbb{K}$, a polynomial $p(x) \in \mathbb{K}[x]$ such that $2 \leq \deg p \leq 3$, is irreducible if and only if it has no roots in $\mathbb{K}$.*

Lemma 1.17 (Gauss). *Suppose $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{Z}[x]$ is primitive, that is $\gcd(a_0, a_1, \dots, a_n) = 1$. Then $p(x)$ is irreducible in $\mathbb{Z}[x]$ if and only if it is irreducible in $\mathbb{Q}[x]$.*

Lemma 1.18 (Eisenstein's Criterion). *Let $q(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{Z}[x]$ and suppose there exists a prime $p \in \mathbb{P}$ such that:*

1. $p \nmid a_n$
2. $p \mid a_0, \dots, a_{n-1}$
3. $p^2 \nmid a_0$

Then $q(x)$ is irreducible in $\mathbb{Q}[x]$.

Example 1.19.

1. The polynomial

$$q(x) = 2x^3 + 6x^2 - 9x + 15$$

satisfies Eisenstein's criterion with $p = 3$.

2. The polynomial

$$q(x) = x^2 + x + 1$$

does not satisfy Eisenstein's criterion. Nevertheless, one can consider the transformation

$$r(x) = q(x+1) = (x+1)^2 + (x+1) + 1 = x^2 + 3x + 3,$$

which does satisfy the criterion. Hence $q(x)$ is also irreducible.

More generally, using the same method as above, one can show that for any prime $p \in \mathbb{P}$, the polynomial

$$q(x) = x^{p-1} + x^{p-2} + \dots + 1$$

is irreducible.